A Weibull counterexample to a mean-normalized tail inequality

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Abstract

Steinerberger asked whether every absolutely continuous, nonnegative, unbounded distribution with finite mean μ admits a threshold $z > 0$ for which the ratio of the probabilities of $\{X+Y \geq 2z,\ \min(X,Y) \leq z\}$ and $\{\min(X,Y) \geq z\}$ is at least $2z/\mu$, for independent copies X, Y. We disprove this assertion using the Weibull survival function $\mathbb{P}(X \geq x) = e^{-x^{3/2}}$. A Gaussian majorant and a one-variable maximization argument give a strict uniform bound below the proposed constant. The proof uses three explicitly specified finite rational checks, with a short exact-arithmetic verifier.

1 The counterexample

For independent copies of a nonnegative random variable X, put

$$R_X(z) = \frac{\mathbb{P}(X+Y \geq 2z,\ \min(X,Y) \leq z)}{\mathbb{P}(\min(X,Y) \geq z)}, \qquad z > 0.$$

Steinerberger's mean-normalized question in [1, Section 1.3] asks whether some $z > 0$ always satisfies $R_X(z) \geq 2z/\mathbb{E}X$. An exponential distribution gives equality at every threshold. The following distribution shows that this behavior is not extremal for the existence question.

Theorem 1. *Let X, Y be independent with density*

$$f(x) = \frac{3}{2}\sqrt{x}\, e^{-x^{3/2}}, \qquad x \geq 0.$$

Their common mean is $\mu = \Gamma(5/3)$. For every $z > 0$,

$$\frac{\mu R_X(z)}{2z} \leq c^{1/3} < 1, \qquad c = \frac{7381964831571}{7530880000000}.$$

In particular, the proposed mean-normalized inequality fails at every positive threshold for this absolutely continuous distribution of unbounded support.

This resolves the mean version of the question in [1].

2 Reduction to a Gaussian integral

Write $S(x) = e^{-x^{3/2}}$ and

$$I(z) = \int_0^z f(x) S(2z-x)\, dx.$$

The two events obtained by designating which variable is at most z are disjoint up to a null event. Independence therefore gives

$$R_X(z) = \frac{2I(z)}{S(z)^2}, \qquad Q(z) := \frac{\mu R_X(z)}{2z} = \frac{\mu I(z)}{zS(z)^2}.$$

The substitution $x = z(1-u)$ yields

$$Q(z) = \frac{3}{2}\mu\sqrt{z}\int_0^1 \sqrt{1-u}\,\exp\Big(-z^{3/2}\big[(1-u)^{3/2}+(1+u)^{3/2}-2\big]\Big)\,du. \tag{1}$$

For $0 \le u \le 1$,

$$(1-u)^{3/2}+(1+u)^{3/2}-2 \ge \frac{3}{4}u^2, \qquad \sqrt{1-u} \le 1-\frac{u}{2}. \tag{2}$$

Indeed, the second derivative of the expression on the left in the first inequality is $\frac{3}{4}((1-u)^{-1/2}+(1+u)^{-1/2}) \ge \frac{3}{2}$ on $[0,1)$; its value and first derivative at zero vanish. Continuity supplies the endpoint. The other inequality follows by concavity of the square root.

Set

$$t = \frac{3}{4}z^{3/2}, \qquad F(t) = \int_0^1 (1-u/2)e^{-tu^2}\,du, \qquad H(t) = t^{1/3}F(t).$$

Equations (1)–(2) imply

$$Q(z) \le \frac{3}{2}\mu(4/3)^{1/3}H(t). \tag{3}$$

Substituting $r = x^{3/2}$ in the mean integral gives $\mu = \Gamma(5/3)$. Hölder's inequality gives

$$\Gamma(5/3) \le \Gamma(3/2)^{2/3}\Gamma(2)^{1/3}, \qquad \mu^3 \le \frac{\pi}{4}.$$

Consequently,

$$Q(z)^3 \le \frac{9\pi}{8}H(t)^3 < \frac{99}{28}H(t)^3, \tag{4}$$

using $\pi < 22/7$.

3 A uniform bound for the integral

Lemma 1. *The function H has a unique maximum at some $t_* > 41/20$. At its maximum,*

$$H(t_*) = G(t_*), \qquad G(t) = \frac{3}{4}t^{1/3}\left[(2-t^{-1})e^{-t}+t^{-1}\right].$$

Moreover, G is strictly decreasing for $t \ge 2$.

Proof. Let $M(t) = \int_0^1 u^2(1-u/2)e^{-tu^2}\,du$. Then

$$H'(t) = \frac{1}{3}t^{-2/3}[F(t) - 3tM(t)].$$

Consider the probability density proportional to

$$e^{-s^2}\left(1-\frac{s}{2\sqrt{t}}\right)\mathbf{1}_{[0,\sqrt{t}]}(s).$$

Its second moment is $q(t) = tM(t)/F(t)$. If $t_2 > t_1$, the ratio of the two unnormalized densities on their common support is

$$\frac{1-s/(2\sqrt{t_2})}{1-s/(2\sqrt{t_1})},$$

which is strictly increasing in s; the larger parameter also adds positive density to the right of the old support. After normalization the density difference has one change of sign, from

negative to positive. Integrating against the strictly increasing function s^2 shows that $q(t)$ is strictly increasing. This sign-change argument also applies if a density ratio never crosses one on the common support, since the total integral of the difference is zero and the additional support lies to the right.

As $t \downarrow 0$, $q(t) \to 0$. As $t \to \infty$, dominated convergence after the substitution $s = \sqrt{t}\,u$ gives $q(t) \to 1/2$, the second moment of the normalized density e^{-s^2} on $[0, \infty)$. Thus $q(t) = 1/3$ has exactly one solution t_*. The derivative formula shows that it is the unique maximum of H.

Put $a = 41/20$. The finite rational inequalities proved below give $F(a) - 3aM(a) > 0$, so $t_* > a$. For $A(t) = \int_0^1 e^{-tu^2}\,du$, elementary integration by parts gives

$$F(t) = A(t) - \frac{1 - e^{-t}}{4t}, \quad M(t) = \frac{A(t) - e^{-t}}{2t} - \frac{1 - (t+1)e^{-t}}{4t^2}.$$

Substituting $F(t_*) = 3t_*M(t_*)$ yields the asserted formula for G. Finally,

$$G'(t) = \frac{3}{4}t^{-5/3}\left[-\frac{2}{3} + e^{-t}\left(-2t^2 + \frac{5}{3}t + \frac{2}{3}\right)\right] < 0 \quad (t \geq 2).$$

The quadratic in parentheses is negative at 2 and decreases thereafter. □

For completeness, all finite numerical constants in the argument are certified by the following rational expressions. Define

$$U_m(x) = \sum_{k=0}^{m} \frac{(-x)^k}{k!}\left(\frac{1}{2k+1} - \frac{1}{4k+4}\right),$$

$$V_m(x) = \sum_{k=0}^{m} \frac{(-x)^k}{k!}\left(\frac{1}{2k+3} - \frac{1}{4k+8}\right).$$

Taylor's theorem with its signed remainder gives lower bounds for e^{-x} at odd degree and upper bounds at even degree, for every $x \geq 0$. Hence

$$F(a) - 3aM(a) \geq U_5(a) - 3aV_6(a).$$

Direct rational arithmetic, reproduced in the appendix, gives

$$U_5(a) > \frac{241}{500}, \qquad V_6(a) < \frac{39}{500}, \qquad \sum_{k=0}^{6} \frac{a^k}{k!} > \frac{100}{13}. \tag{5}$$

Thus $F(a) - 3aM(a) > 23/10000 > 0$, and the last inequality, together with the positive Taylor series for e^a, gives $e^{-a} < 13/100$. By Lemma 1, $H(t) \leq G(t_*) < G(a)$ for every $t > 0$. Since $2 - a^{-1} > 0$, equations (4) and (5) now imply

$$Q(z)^3 < \frac{99}{28}\frac{41}{20}\left[\frac{3}{4}\left(\frac{62}{41}\frac{13}{100} + \frac{20}{41}\right)\right]^3 = \frac{7381964831571}{7530880000000} < 1.$$

This proves Theorem 1 for all $z > 0$.

A Exact verification of the finite inequalities

The following Python 3 code uses rational arithmetic only. The functions `factorial` and `Fraction` are in the standard library. The positive gap in the first check is a lower bound, not an approximation of the integral. The preceding proof supplies the estimates on the entire continuous range $z > 0$.

```
from fractions import Fraction as Q
from math import factorial
a = Q(41, 20)
def U(m):
    return sum((-a)**k / factorial(k) *
               (Q(1,2*k+1)-Q(1,4*k+4)) for k in range(m+1))
def V(m):
    return sum((-a)**k / factorial(k) *
               (Q(1,2*k+3)-Q(1,4*k+8)) for k in range(m+1))
assert U(5) > Q(241,500)
assert V(6) < Q(39,500)
assert sum(a**k/factorial(k) for k in range(7)) > Q(100,13)
c = Q(99,28)*a*(Q(3,4)*((2-1/a)*Q(13,100)+1/a))**3
assert c == Q(7381964831571,7530880000000)
assert c < 1
```

The accompanying reproducibility directory contains an independently written verifier and its exact rational output.

References

[1] S. Steinerberger, *A sharp estimate for probability distributions*, Statistics & Probability Letters **155** (2019), 108584. doi:10.1016/j.spl.2019.108584. Preprint: arXiv:1801.00663, Section 1.3.